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Multiaxial Finite Strain Behavior of Polydomain Liquid Crystal Elastomers

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Abstract

Liquid Crystal Elastomers (LCEs) represent a class of stimuli-responsive polymers that combine the order-dependent properties of liquid crystal mesogens with the soft elasticity of polymer networks. Apart from some benchmark loading scenarios such as uniaxial loading tests, the literature lacks a theoretical framework addressing the response of LCEs under complex, multiaxial loading. For this purpose, this study investigates the stress response of polydomain LCE cylinders and balloons under loading cases including pure extension–torsion, inflation, and extension–torsion–inflation. To do so, a robust analytical framework was developed by assuming polydomain LCEs to be isotropic hyperelastic materials in order to alleviate the need for step-length tensors and anisotropy parameters in purely mechanical loadings. The analytical framework was validated using finite element method (FEM) results, revealing excellent agreement between predictions. To assess the stress response of the hyperelastic LCEs considered, the analytical framework employed three different forms of the strain energy function, namely Mooney–Rivlin, Yeoh, and Anssari-Benam. Among them, the Yeoh and Anssari-Benam models were capable of capturing inherent strain-softening and large-deformation effects in LCEs across a wide range of stretch values. This study also shows that strain energy functions of binomial form, such as Anssari-Benam, provide the most accurate predictions for all stress components. As practical considerations, the study derived the blocking force and the resultant longitudinal torque responses of polydomain LCEs, and illustrated the through-thickness instability phenomenon occurring in LCE balloons with respect to different geometric factors. These results offer a straightforward pathway for the analysis and design of polydomain LCEs that are mainly deployed as soft actuators and peristaltic pumps.

Keywords: Polydomain Liquid Crystal Elastomers, Thick-walled Structures, Large Deformations, Multiaxial Loading, Soft Actuators.

1. Introduction

Liquid Crystal Elastomers (LCEs) are crosslinked polymer networks that incorporate rigid, rod-like molecules called mesogens, either in the main chain or as side groups. These mesogens impart anisotropy (directional order) to the material, while the elastomer network provides soft elasticity. LCEs are also responsive to thermal stimuli, where the mesogen alignment and the corresponding degree of anisotropy change significantly with temperature. Above a threshold known as the nematic-to-isotropic transition temperature (T_{ni}), the LCE microstructure reaches an isotropic state. Conversely, below this temperature, the mesogens are aligned along a preferred direction referred to as the nematic state. This smart, temperature-dependent behavior makes LCEs suitable candidates for a wide range of applications, including soft actuators, shape-morphing structures, and biomedical devices [1], [2], [3], [4], [5], [6]. This is particularly notable because LCEs can be reprogrammed to respond to other external stimuli beyond temperature, such as stress, light, electricity and more [7], [8], [9], [10], [11]. More precisely, LCEs can be categorized into three groups based on their mechanical response and fabrication method. The first class are called nematic monodomain LCEs, in which the mesogens are uniformly aligned throughout the entire sample; consequently, the material response can be characterized by a single global director. The second class includes isotropic-genesis polydomain LCEs, where the sample is crosslinked while the mesogens are in the

isotropic phase. Finally, the third class comprises nematic-genesis polydomain LCEs, which are crosslinked in the nematic phase. In this case, the mesogens are organized into numerous small domains with local directors at the mesoscale; however, the collection of these domains is randomly oriented, resulting in macroscopically isotropic behavior [8], [12], [13], [14].

Regarding their mechanical properties, LCEs exhibit a range of complex behaviors, including highly nonlinear stress–strain responses, auxetic behavior, instabilities, and discontinuous softening [15], [16], [17]. These complex responses make the mechanical behavior of LCEs significantly different from that of standard rubber-like materials, which requires a detailed investigation from various perspectives. Analytically, researchers have studied the indentation of LCEs, showing a clear deviation from classical Hertz theory due to soft elasticity. The relationship between adhesion and chain alignment has also been established through detailed analyses, showing that adhesion is stronger at higher nematic orders (i.e., lower temperatures) [18], [19]. For instance, Ohzono et al. found that adhesion in LCE specimens is more than twice as high in the nematic phase as it is in the isotropic state [20]. Additionally, the onset of nonlinear elastic wrinkling and creasing instabilities in LCEs has been examined in several studies. These works show that the degree of anisotropy significantly affects the critical stress and strain required for the onset of instability [17], [21]. The time-dependent and strain rate-dependent behavior of LCEs is another area that has been widely investigated [22]. Consequently, theories have been developed to describe the finite-deformation viscoelastic phenomena observed in LCEs [23], [24], [25]. In terms of structural applications, the energy absorption density has been studied for LCE-based metamaterials, and a power-law relationship was suggested for energy absorption with respect to the strain rate [26]. This relationship can be tuned by adjusting the parameter representing the degree of mesogen alignment.

For the constitutive modeling of LCEs, the hyperelasticity framework is considered the most fundamental starting point. Among the various models available, the neo-classical theory is widely adopted. This theory accounts for the coupling between the soft elasticity of the polymer backbone and the anisotropy induced by the liquid crystal director. This coupling is incorporated into the model by adding two specific terms to the material's strain energy density: an anisotropy parameter (r) and a step-length tensor ($\mathcal{L}$) [27], [28]. Early implementations of neo-classical theory utilized a Neo-Hookean strain energy function, which assumes a Gaussian molecular network. However, later adaptations have employed more advanced models—such as the Mooney-Rivlin [29] and Ogden [30] frameworks—to accurately capture larger deformation regimes. More recent developments have expanded upon these foundations by including the effects of external stimuli, such as thermomechanical and photomechanical couplings [31], [32], [33]. Notably, a recent study provided new insights into modeling LCEs exhibiting an isotropic response, such as isotropic-genesis and nematic-genesis polydomain LCEs. This work demonstrated that these particular LCEs can be treated as conventional hyperelastic materials, bypassing the need for a step-length tensor or an anisotropy parameter [34], [35]. Furthermore, the study proposed a robust strain energy function, so-called the Anssari-Benam model, which captures the complex strain-softening effects inherent to polydomain LCEs and accurately describes their behavior under large deformations.

The development of constitutive models for soft elastic solids has traditionally focused on key benchmarks—such as uniaxial tension, equi-biaxial tension, and pure shear—due to their procedural simplicity and the widespread availability of experimental data. In practical applications, however, soft materials and biological tissues often experience complex multiaxial deformations. For example, deformations in ligaments, tendons, and cardiac tissues typically involve simultaneous shear, axial stretch, and rotation. Consequently, more complex deformation regimes, such as inflation and combined extension–torsion, are addressed in analytical studies to better characterize the behavior of smart materials under realistic conditions [36], [37], [38], [39]. Among different smart materials, the practical applications of LCEs involving complex actuation modes—such as bending, twisting, and rolling—have been explored in several studies [40], [41]. Therefore, investigating the response of LCEs under multiaxial loading scenarios is essential. For instance, analytical solutions for the inflation of LCE balloons have been recently developed [42].

This study examines the finite-strain behavior of polydomain Liquid Crystal Elastomers (LCEs) with thick-walled geometries under multiaxial loading conditions. The loading scenarios considered include pure extension–torsion, inflation, and combined extension–torsion superimposed with inflation, applied to both cylindrical and balloon-like configurations. Section 2 addresses the development of the theoretical framework for

polydomain LCEs, wherein the governing equations for cylindrical and spherical geometries are derived based on their respective boundary conditions. In Section 3, the proposed analytical framework is validated using numerical simulations performed via the finite element method (FEM). Furthermore, the stress responses of polydomain LCEs are analyzed using different constitutive models to evaluate their predictive capabilities. The substantial potential of LCEs under multiaxial loading is emphasized through the assessment of blocking forces and resultant torques at actuation states, illustrating their relevance in soft actuator applications. Finally, Section 4 presents the concluding remarks along with recommendations for potential future research directions concerning polydomain LCEs.

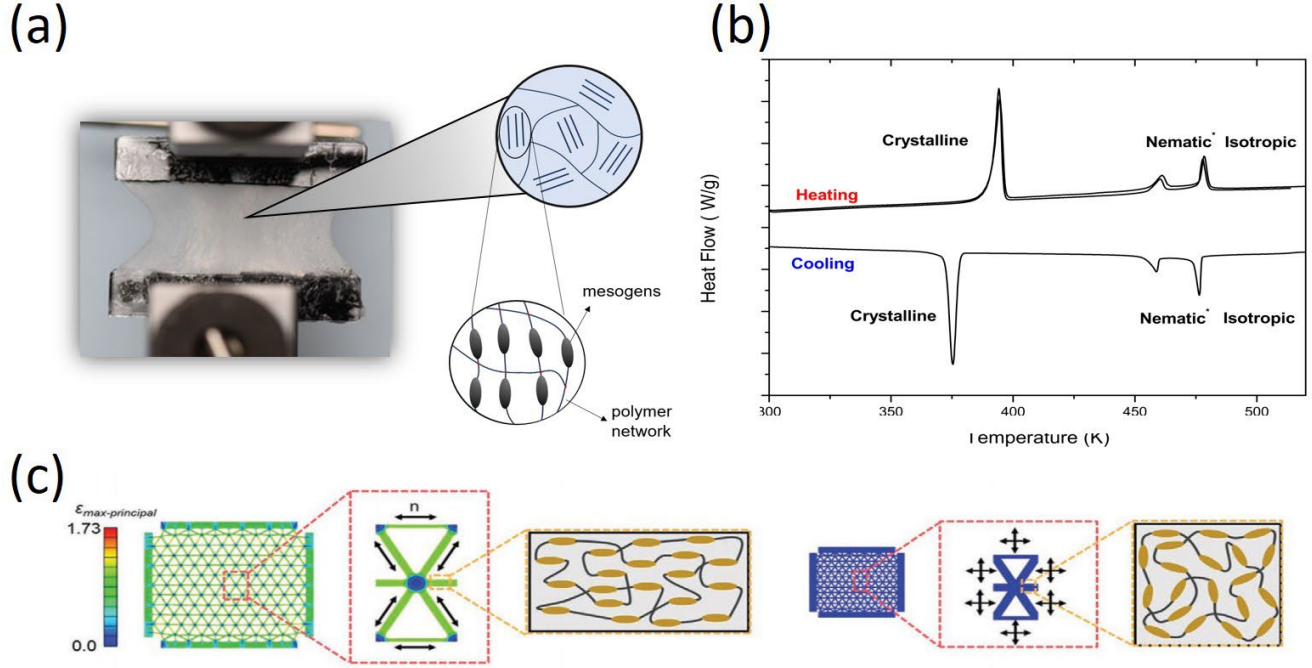


Figure 1. Structural and thermal representation of Liquid Crystal Elastomers (LCE). a) Schematic of a polydomain LCE specimen across multiple scales. The macroscopic specimen is shown under tensile loading, revealing a microstructure composed of randomly oriented domains, each consisting of mesogens aligned along a local director (macroscopic specimen image adapted from [34]). b) Representative differential scanning calorimetry thermogram illustrating the crystalline, smectic, and nematic phase transitions in a liquid crystal. While enthalpy peaks remain largely consistent for both monodomain and polydomain LCEs, polydomain peaks may appear broader or slightly shifted due to the lack of global alignment (Adapted from [43]). c) Multiscale design of LCE metamaterials, contrasting the molecular alignment in monodomain (left) and polydomain (right) lattice architectures (Adapted from [6]).

2. Theoretical Framework

To investigate the finite-strain behavior of polydomain LCE structures at the equilibrium state, a theoretical framework is required that provides a closed-form solution for the problem. A numerical FEM analysis is also required for the verification of the results. COMSOL Multiphysics® v.6.3 software is used for both aforementioned frameworks. In the theoretical framework, the governing partial differential equations are established and solved directly. The following sections detail the development of such a framework for the boundary-valued problems of polydomain LCE cylinders and balloons under multiaxial loading, respectively.

2.1. LCE cylinder Governing Equations

The constitutive model development for the extension-torsion of a thick-walled cylinder begins by defining the deformation mapping $\chi(X)$ from the reference configuration (R, θ, Z) to the current configuration (r, θ, z) in a cylindrical coordinate system as follows:

$$r = r(R), \quad \theta = \theta + r\tau Z, \quad z = \lambda Z. \quad (1)$$

where τ denotes the twisting angle per unit length, and λ represents the uniform axial stretch. The cylinder is assumed to be of infinite length. In Eq. (1), the unknown function $r(R)$ is obtained from the displacement field. Hereafter, in the governing equations, upper-case letters refer to the stress-free reference configuration, while lower-case letters denote the current configuration. With respect to the displacement-field assumptions, the cross-sections may undergo rigid translation along the Z -axis and rigid rotation about the longitudinal axis. The cross-sections remain flat and orthogonal to the longitudinal axis, preserving axial symmetry and the circular geometry. Moreover, the torsion is assumed to vary linearly along the longitudinal axis of the cylinder. The kinematic relationship between these configurations is characterized by the deformation gradient tensor, $\mathbf{F}$, which represents the local linearization of the mapping. For the deformation described in Eq. (1), the components of $\mathbf{F}$ are derived as:

$$\mathbf{F} = \begin{bmatrix} \frac{\partial r}{\partial R} & 0 & 0 \\ 0 & \frac{r}{R} & r\tau \\ 0 & 0 & \lambda \end{bmatrix} \quad (2)$$

The deformation gradient $\mathbf{F}$ is not frame-indifferent, as it conflates rigid-body rotations with material stretching and therefore does not serve as a pure measure of strain. To isolate the material deformation from these rotations, the Cauchy–Green deformation tensors are commonly employed. In this study, the left Cauchy–Green tensor is adopted:

$$\mathbf{B} = \mathbf{F}\mathbf{F}^T = \begin{bmatrix} \left(\frac{\partial r}{\partial R}\right)^2 & 0 & 0 \\ 0 & \left(\frac{r}{R}\right)^2 + (r\tau)^2 & r\tau\lambda \\ 0 & r\tau\lambda & \lambda^2 \end{bmatrix} \quad (3)$$

With the kinematics of the problem established, we now characterize the kinetics governing the stress state within the material. The constitutive model adopted for polydomain LCEs is formulated within a hyperelastic framework, assuming isothermal conditions and neglecting dissipative effects. Under these assumptions, the material response is described by a strain energy density function, which represents the stored energy as a function of the deformation measure.

We employ the recently developed strain energy function proposed by Ansari–Benam et al. [34]. Compared with classical formulations, such as the Neo-Hookean model, this binomial strain energy function demonstrates a strong capability to capture finite deformations and the complex mechanical behavior of a wide range of isotropic soft materials. The authors showed that their model can be considered a parent to most invariant-based hyperelastic models in the literature, accurately describing the response of both isotropic-genesis and nematic-genesis polydomain LCEs under benchmark loading conditions. Furthermore, the proposed model satisfies the polyconvexity criterion over a wide range of deformations (see Fig. 2), thereby ensuring mathematical stability. The Ansari–Benam strain energy function is defined as:

$$W(\bar{I}_1, \bar{I}_2) = \sum_{j=1} \left(3 \frac{n_j - 1}{2n_j} \right) \mu_j N_j \left[\left(\frac{1}{3N_j(n_j - 1)} \right) (I_1 - 3)^{\beta_j} - \ln \left(\frac{I_1 - 3N_j}{3 - 3N_j} \right)^{\beta_j} \right] + \sum_{k=1} C_k \left[\left(\left(\frac{I_2}{3} \right)^{\varepsilon_k} \right) - 1 \right] \quad (4)$$

where $n_j, \mu_j, N_j, \beta_j, C_k, \varepsilon_k$ are material parameters satisfying:

$$\{n_j, \mu_j, N_j, C_k \in \mathbb{R}^+, \beta_j, \varepsilon_k \in \mathbb{R}\} \quad (5)$$

Similar to most constitutive models in the literature, the Ansari–Benam strain energy function (see Eq. (4)) is formulated using the first and second invariants of the Cauchy–Green deformation tensors. It should be noted that

the Anssari–Benam model assumes material incompressibility, implying that the ratio of the current volume to the initial volume remains unity. Equivalently, the third invariant of the Cauchy–Green tensors is defined as $I_3 = 1$ (or $J = \det(F) = 1$). This assumption ensures that the deviatoric components are the sole contributors to the total invariants of the left Cauchy–Green deformation tensor, as the volumetric parts vanish (i.e., $I_1 = J^{\frac{-2}{3}} \bar{I}_1 = \bar{I}_1$, $I_2 = J^{\frac{-4}{3}} \bar{I}_2 = \bar{I}_2$). Consequently, the invariants are defined as:

$$I_1 = \text{tr}(\mathbf{B}), \text{ and } I_2 = \frac{1}{2}[(\text{tr} \mathbf{B})^2 - \text{tr}(\mathbf{B}^2)]. \quad (6)$$

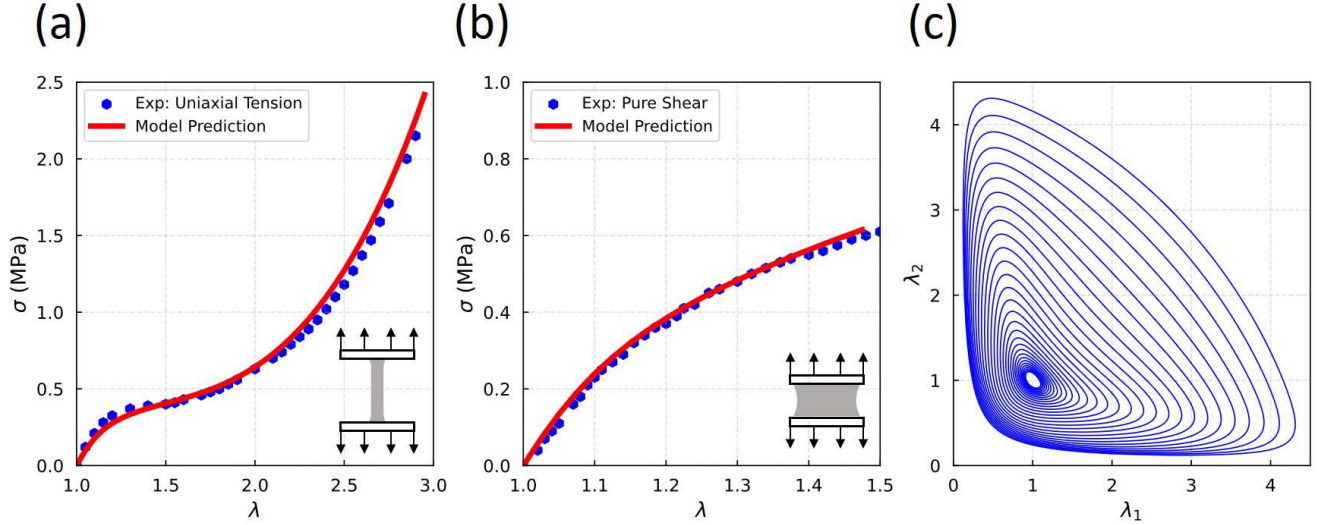


Figure 2. Single-term expansion of the Anssari–Benam binomial strain energy function, fitted to experimental data of nematic-genesis polydomain LCEs subjected to (a) uniaxial tension and (b) pure shear. (c) Iso-energy contours for the single-term expansion, illustrating its polyconvex behavior over a wide range of deformations.

It is demonstrated that a single-term expansion of the Anssari–Benam strain energy function (Eq. (4)) is sufficient to simultaneously capture the response of nematic-genesis polydomain LCEs under both pure shear and uniaxial tension (see Fig. 2). For this case, the general formulation of the Anssari–Benam model reduces to:

$$W(\bar{I}_1, \bar{I}_2) = 3 \frac{n_1 - 1}{2n_1} \mu_1 N_1 \left[\left(\frac{1}{3N_1(n_1 - 1)} \right) (I_1 - 3)^{\beta_1} - \ln \left(\frac{I_1 - 3N_1}{3 - 3N_1} \right)^{\beta_1} \right] + C_{20} \left[\left(\left(\frac{I_2}{3} \right)^{\varepsilon_1} \right) - 1 \right] \quad (7)$$

which μ_1 , C_{20} , n_1 , N_1 , β_1 , and ε_1 are the material constants specific to the single-term expansion. The material parameters obtained from the fitting procedure are summarized in Table 1.

Table 1. Material parameters obtained for the single-term expansion of the Anssari–Benam strain energy function.

μ_1 (MPa)	C_{20} [MPa]	n_1 [-]	N_1 [-]	β_1 [-]	ε_1 [-]
0.02	2.87	0.85	0.965	2.39	0.28

With the kinematic description and strain energy density function established, the Cauchy stress tensor is derived within the framework of hyperelasticity in terms of the deformation measures and the derivatives of the strain energy function as

$$\boldsymbol{\sigma} = -p\mathbf{I} + 2W_1\mathbf{B} - 2W_2\mathbf{B}^{-1} \quad (8)$$

which $\mathbf{I}$ is the identity tensor and p is a Lagrange multiplier enforcing the incompressibility constraint, evaluated according to the specific boundary and internal conditions of each problem. The quantities W_1 and W_2 denote the partial derivatives of the strain energy density function W with respect to the first and second invariants, I_1 and I_2 , respectively, i.e., $W_1 = \frac{\partial W}{\partial I_1}$ and $W_2 = \frac{\partial W}{\partial I_2}$. The non-zero components of the Cauchy stress tensor are derived in Appendix A.

In addition to the Cauchy stress tensor, the first Piola–Kirchhoff stress tensor $\mathbf{P}$ is another important stress measure used to fit the experimental data to the proposed constitutive model. The first Piola–Kirchhoff stress tensor is related to the Cauchy stress tensor by:

$$\mathbf{P} = J \boldsymbol{\sigma} \mathbf{F}^{-T} \quad (9)$$

For the incompressible case considered here ($J = 1$), Eq. (9) reduces to $\mathbf{P} = \boldsymbol{\sigma} \mathbf{F}^{-T}$.

The final step in defining the problem is imposing the equilibrium state. At equilibrium, the stress and strain states of the continuum are governed by the principle of minimum total potential energy. For a quasistatic regime in the absence of body forces, the conservation of linear momentum reduces to the equations of equilibrium. For a thick-walled hollow cylinder, the radial equilibrium of Cauchy stress components in cylindrical coordinates is expressed as:

$$\frac{\partial \sigma_{rr}}{\partial r} + \frac{\sigma_{rr} - \sigma_{\theta\theta}}{r} = 0 \quad (10)$$

The boundary-value problem is fully specified by prescribing the appropriate boundary conditions, defined as:

$$\sigma_{rr}|_{r=r_i} = -p_i, \text{ and } \sigma_{rr}|_{r=r_o} = 0. \quad (11)$$

These boundary conditions imply that the inner surface ($r = r_i$) is subjected to an internal pressure p_i during combined extension–torsion and inflation, whereas it is traction-free ($p_i = 0$) under pure extension–torsion. The outer surface ($r = r_o$) is assumed to be traction-free in all cases.

2.2. LCE balloon Governing Equations

Following the establishment of the analytical framework for LCE cylinders, we now present the governing equations for the inflation of a thick-walled balloon composed of polydomain LCEs. Compared to the governing equations for LCE cylinders established in Section 2.1, the differences in the governing equations for thick-walled balloons lie in the kinematics, equilibrium equation, and boundary conditions. For a thick-walled balloon under inflation, a spherical coordinate system is most appropriate. Consequently, we employ this system to develop the kinematics of the problem, utilizing a deformation mapping described in spherical coordinates. Schematics of the LCE balloon in the spherical coordinate system, along with a cross-sectional view in the x-y plane before and after applying the inner pressure p_i , are shown in Figure 3a. By applying the inner pressure to a thick-walled balloon, a material point in the stress-free reference configuration, denoted by (R, Θ, Φ) , moves to the current configuration (r, θ, φ) via:

$$r = r(R), \quad \theta = \Theta, \quad \varphi = \Phi. \quad (12)$$

This mapping yields the following explicit form for the deformation gradient, $\mathbf{F}$:

$$\mathbf{F} = \text{diag}\left(\frac{\partial r}{\partial R}, \frac{r}{R}, \frac{r}{R}\right) \quad (13)$$

Consistent with the previous case study, the left Cauchy–Green tensor, $\mathbf{B}$, is used to measure rotation-free deformations in the thick-walled LCE balloon at hand. Moreover, the incompressibility condition is imposed, implying:

$$J = \left(\frac{\partial r}{\partial R}\right) \left(\frac{r}{R}\right)^2 = 1 \quad (14)$$

Specializing to the deformation field of Eq. (13), the only remaining equilibrium equation is:

$$\frac{\partial \sigma_{rr}}{\partial r} + \frac{2(\sigma_{rr} - \sigma_{\theta\theta})}{r} = 0 \quad (15)$$

Eq. (15) governs the equilibrium state along the radial direction of the balloon. Notably, because we are analyzing a thick-walled geometry, the radial stress σ_{rr} cannot be neglected. Symmetry dictates that the hoop stress components are equal ($\sigma_{\theta\theta} = \sigma_{\phi\phi}$), which simplifies the equilibrium equation accordingly.

The boundary condition corresponding to an internal pressure p_i acting on the balloon's interior surface is:

$$\sigma_{rr}|_{r=r_i} = -p_i, \quad \sigma_{rr}|_{r=r_o} = 0. \quad (16)$$

As indicated in Eq. (18), the inner surface ($r = r_i$) is subjected to pressure p_i , while the outer surface ($r = r_o$) is traction-free.

3. Results and Discussion

3.1. Cylinder response

This section discusses the mechanical response of LCE cylinders under pure extension–torsion and extension–torsion–inflation loading scenarios. A schematic of the thick-walled cylinder configuration before and after the deformation mapping is shown in Figure 3a. Note that the length of the cylinder along the z-axis does not carry physical meaning, as the cylinder is assumed to be of infinite length. As mentioned earlier, the problem is addressed in two ways: (i) by analytically solving the boundary-value problem through formulation of the equilibrium equation, and (ii) by directly implementing a FEM model. In the analytical approach, the expanded equations presented in Section 2.2 are formulated explicitly. The governing partial differential equation (PDE), which depends on the specific case study (Eq. (10) or Eq. (15)), along with the incompressibility condition ($J = 1$), are solved simultaneously using the *PDE Interfaces* module in COMSOL Multiphysics® v6.3.

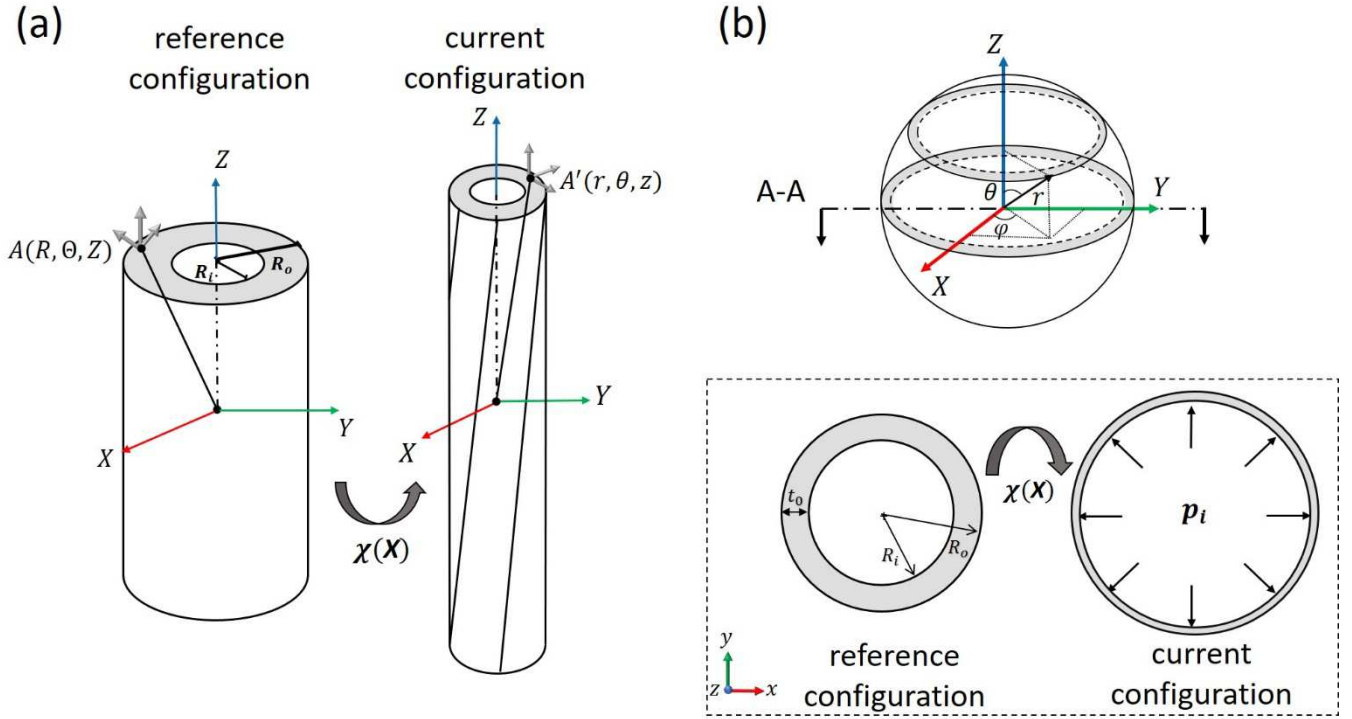


Figure 3. (a) Schematic of the thick-walled LCE cylinder subjected to combined extension–torsion. An arbitrary point A and its corresponding point A' after application of the deformation mapping $\chi(X)$ are shown in the reference and current configurations, respectively. Cartesian and cylindrical coordinate systems, together with their spatial transformations, are illustrated. (b) Schematic of an LCE balloon subjected to an internal pressure p_i . Cross-sectional views in the $x - y$ plane are presented for both the reference and current configurations.

For the direct FEM simulations, the *Solid Mechanics* module of the COMSOL Multiphysics interface was used, and the adopted strain-energy functions were assigned to the hyperelastic material model. Tetrahedral elements with quadratic serendipity shape functions were employed for the spatial discretization of the domain. All simulations were performed in a quasi-static regime, and in most of the conducted studies, the incremental analyses were carried out by continually increasing the value of the corresponding loading variable (twisting angle or stretch). To prevent numerical divergence in the FEM simulations, the boundary conditions were symmetrically applied at both ends of the cylinder. Specifically, half of the prescribed torsion and axial stretch ($\tau/2$ and $\lambda/2$) were imposed on the top end, while the bottom end was assigned the corresponding negative values ($-\tau/2$ and $-\lambda/2$). This symmetric loading strategy stabilizes the numerical solution while producing results equivalent to a configuration in which one end is fully fixed and the full boundary-condition values are applied to the opposite end.

For verification of the numerical and analytical models, the radius changes across the cylinder thickness under extension-torsion loading were compared. Figure 4 shows the FEM results alongside the analytical solution for moderate levels of stretch and twist. Excellent agreement is observed between the two approaches, confirming the robustness of the analytical model assumptions. A linear shrinkage along the thickness is observed in the results for the pure extension-torsion case, which increases as the applied stretch is augmented.

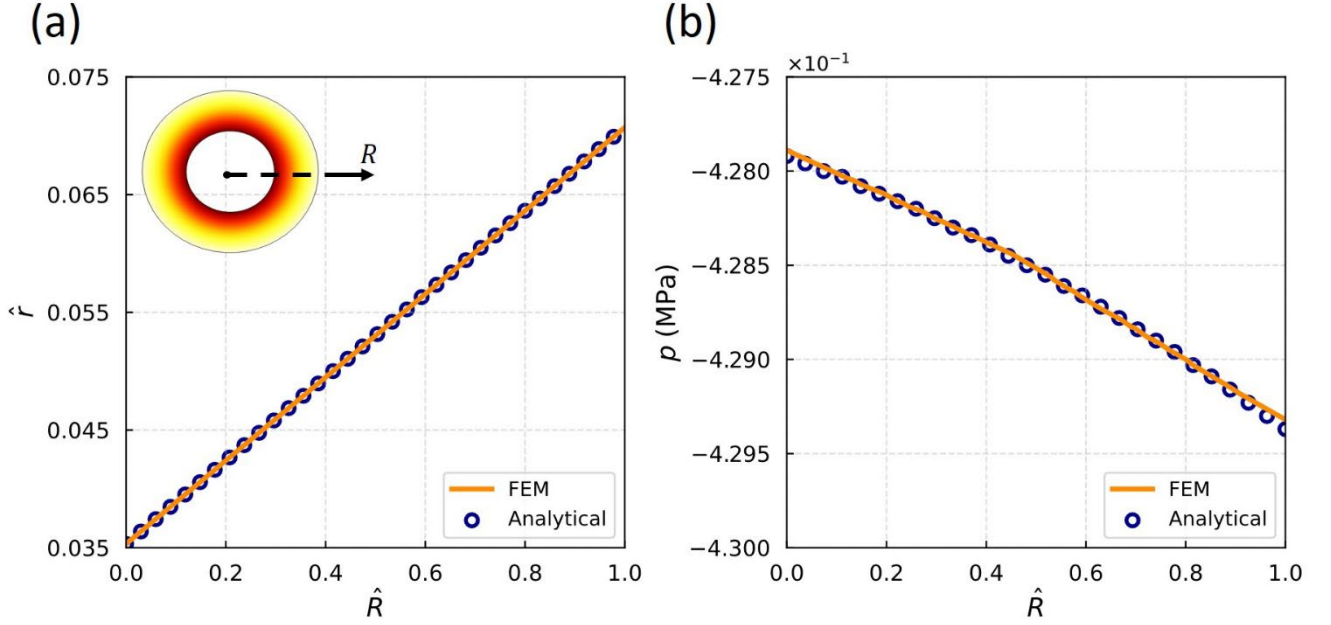


Figure 4. Verification of the analytical solutions against direct FEM results for fixed loading conditions of $\lambda = 2$ and $\tau = \pi/2$. (a) Radial shrinkage of the LCE cylinder across the wall thickness (r-coordinate) following deformation; a schematic of the radial coordinate change is also presented. (b) Variation of the auxiliary pressure p along the wall thickness (arc length). Solid lines represent the direct FEM results, whereas unfilled circles denote the analytical predictions.

Since soft materials such as LCEs frequently undergo large deformations in practical applications, characterizing their mechanical response beyond the small-strain regime is essential. In this deformation range, direct FEM simulations often encounter numerical instabilities and convergence difficulties. To address these challenges, the primary objective of developing closed-form analytical solutions for LCEs was to provide a robust and efficient alternative framework.

After validating the proposed analytical framework against numerical results for isotropic LCEs at moderate stretch levels, we proceed to investigate the stress–stretch response of polydomain LCEs under large deformations. In this context, the proposed model (see Eq. (7)) is examined alongside two well-established constitutive models: the Mooney–Rivlin and Yeoh strain energy functions, defined as follows:

$$W_{\text{Mooney-Rivlin}}(I_1, I_2) = c_{10}(I_1 - 3) + c_{01}(I_2 - 3) \quad (17)$$

$$W_{\text{Yeoh}}(I_1) = c_1(I_1 - 3) + c_2(I_1 - 3)^2 + c_3(I_1 - 3)^3 \quad (18)$$

where c_{10} , c_{01} , c_1 , c_2 , and c_3 are material parameters obtained by calibrating the predicted stresses of each strain energy function against benchmark experimental data, specifically uniaxial tensile tests.

The stress predictions obtained using the Mooney–Rivlin, Yeoh, and Anssari–Benam constitutive models are presented in Figure 5. The calibration results for the Mooney–Rivlin and Yeoh models, expressed in terms of the first Piola–Kirchhoff stress tensor $\mathbf{P}$ (see Eq. (9)), are also shown in this figure, while the identified material parameters are summarized in Table 2. The stresses are evaluated at the outer radius ($R = R_o$) of the LCE cylinder subjected to pure extension–torsion loading. Note that σ_{rr} vanishes throughout the radial domain and is therefore not displayed.

Table 2. Material parameters for the Mooney–Rivlin and Yeoh constitutive models used in this study.

Constitutive Model		Model Parameters (MPa)		
Mooney-Rivlin	$c_{01} = 0.1861$	$c_{10} = 0.0815$	—	
Yeoh	$c_1 = 0.206$	$c_2 = -0.0152$	$c_3 = 0.00335$	

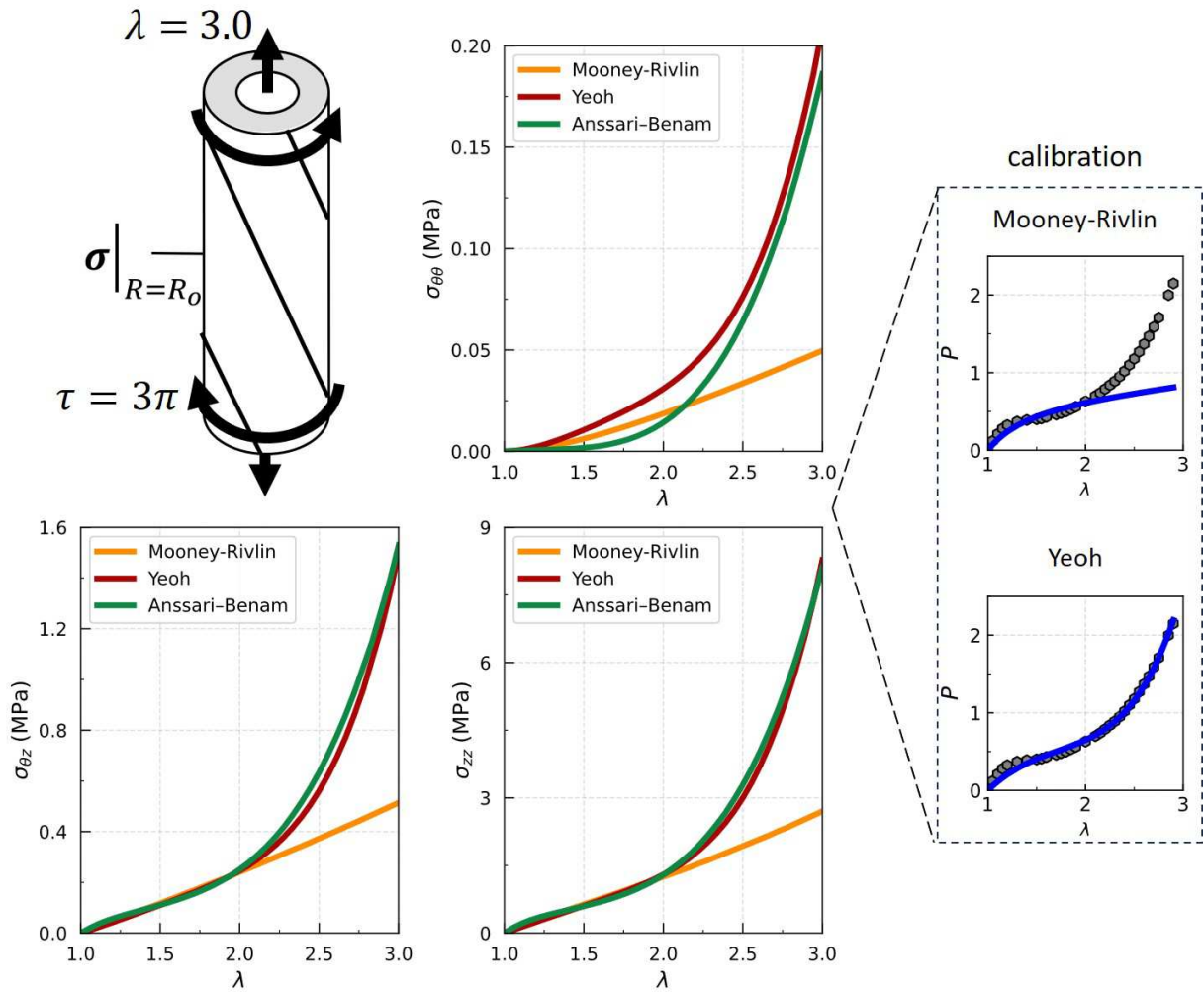


Figure 5. Analytical predictions of the stress–stretch response of LCE cylinders under combined extension–torsion based on different constitutive models. Results are evaluated at the outer surface of the cylinder ($R = R_0$). The twist angle is fixed at $\tau = 3\pi$. The fitting results for the Yeoh and Mooney–Rivlin models, obtained from uniaxial tension data, are presented on the right.

Figure 5 demonstrates that the Anssari–Benam model, formulated as a binomial constitutive model, accurately captures key material features of LCEs, including semi-soft behavior, the non-monotonic stress response observed

at moderate stretch levels, and the extensibility limit. In contrast, although the Mooney–Rivlin model provides reasonable predictions at low to moderate stretch levels, its accuracy deteriorates significantly at higher stretches, leading to a pronounced divergence from the expected response. Moreover, the Mooney–Rivlin model does not capture either the semi-soft behavior of LCEs or the limiting chain extensibility at large stretch levels. As expected, conventional constitutive models such as Mooney–Rivlin are generally unable to represent the complex, non-monotonic mechanical responses characteristic of smart materials. When forced to fit the full range of experimental stretch data, they typically achieve acceptable agreement only within a limited stretch regime (generally $\lambda < 2.0$).

The Yeoh model provides significantly improved predictions compared to the Mooney–Rivlin model. In particular, good agreement is observed between the Yeoh and Anssari–Benam model predictions for the $\sigma_{\theta z}$ and σ_{zz} stress components, which are the primary stresses governing extension–torsion loading (see the lower panels in Figure 5). However, noticeable differences between these two models persist for the remaining stress components, namely σ_{rr} and $\sigma_{\theta\theta}$. To further examine these differences, Figure 6 compares the stress distributions predicted by the Yeoh and Anssari–Benam models along the non-dimensional arc-length coordinate of the cylinder, defined as $\bar{R} = \vec{R} - R_o / R_o - R_i$. The error bars represent the difference between the model predictions, and the maximum relative variation along the arc length is reported in each panel.

Although the predictions exhibit close agreement for the axial and shear driving stresses ($\sigma_{\theta z}$ and σ_{zz}), with relative errors of 13.0% and 1.9%, respectively, larger discrepancies are observed for the radial and circumferential stress components, σ_{rr} and $\sigma_{\theta\theta}$, where the relative errors reach 28.6% and 19.7%, respectively. These differences stem from the distinct constitutive structures of the two models. The Anssari–Benam model incorporates both the first and second invariants of the deformation tensor (I_1 and I_2) in the strain energy function, whereas the Yeoh model depends solely on I_1 . The σ_{rr} and $\sigma_{\theta\theta}$ stress components are strongly influenced by the incompressibility constraint and transverse stress redistribution. From the stress definition (Eq. 8), the inclusion of the second invariant I_2 introduces an additional $\mathbf{B}^{-1}$ -dependent contribution (with $W_2 = \frac{\partial W}{\partial I_2}$) to the Cauchy stress tensor, which enhances multiaxial coupling and affects through-thickness stress gradients. In contrast, the axial and shear stress components exhibit closer agreement because they are primarily governed by the I_1 -dependent term.

Beyond models comparison, Figure 6 also provides insight into the intrinsic through-thickness stress distributions. Under pure extension–torsion, the boundaries are traction-free; therefore, σ_{rr} at the interior and exterior surfaces is zero. Along the arc length, however, σ_{rr} exhibits a second-order-type variation, attaining a minimum approximately at the mid-arc length (see the upper-left panel in Fig. 6). The circumferential and shear stress components, $\sigma_{\theta\theta}$ and $\sigma_{\theta z}$, vary linearly across the cylinder thickness. The circumferential stress distribution is symmetric about the neutral axis of the cross-section, transitioning from negative to positive values through the thickness. By contrast, the axial stress component, σ_{zz} , remains essentially uniform throughout the wall thickness, with negligible variation.

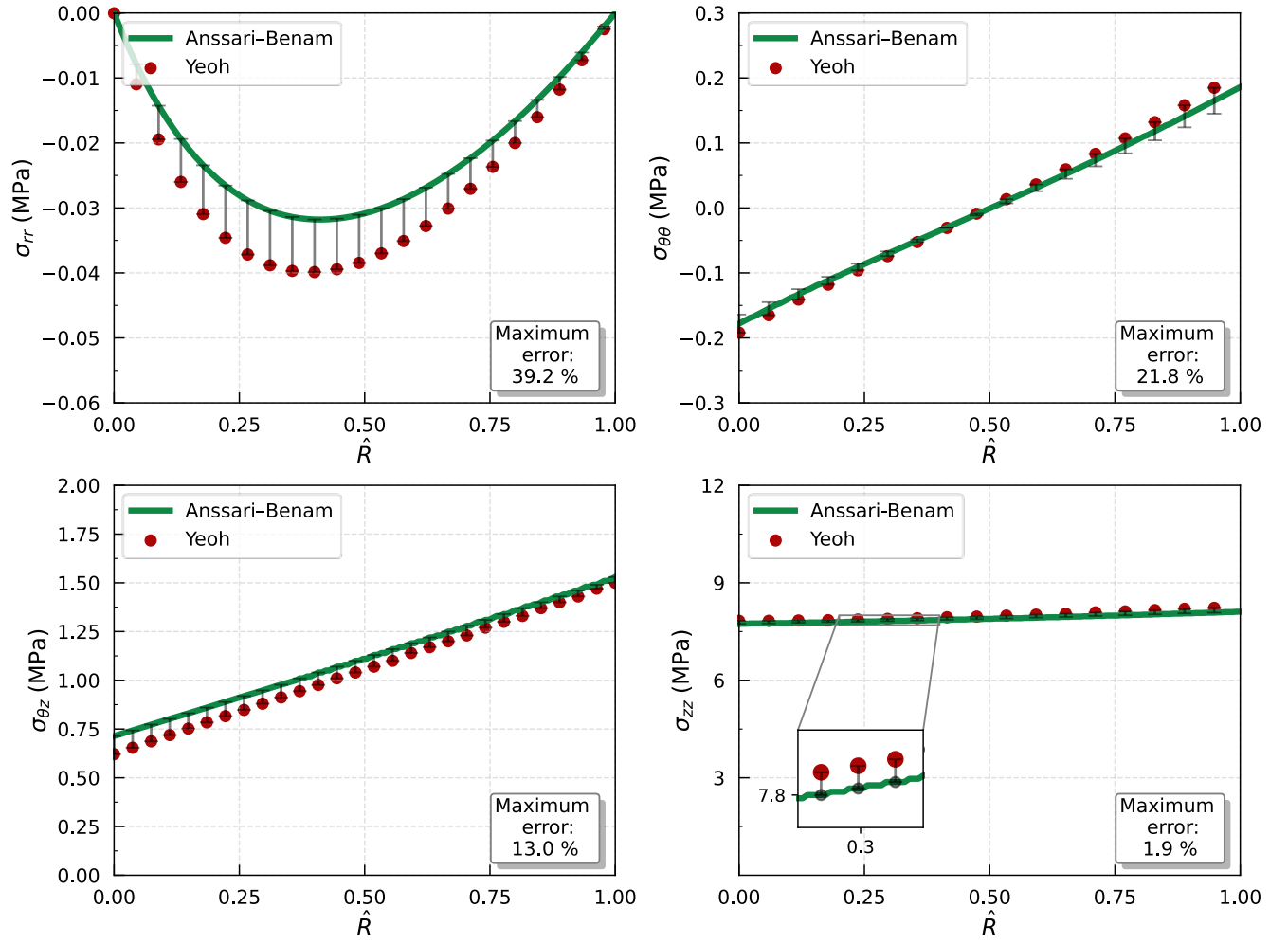


Figure 6. Comparison between the Anssari-Benam and Yeoh model predictions for non-zero stress components, plotted against the non-dimensional arc length of the cylinder. The relative differences between constitutive models' predictions are indicated by error bars, with the maximum error along the entire arc length reported for each case. The kinematic inputs are set as $\tau = 3\pi$ and $\lambda = 3.0$, respectively.

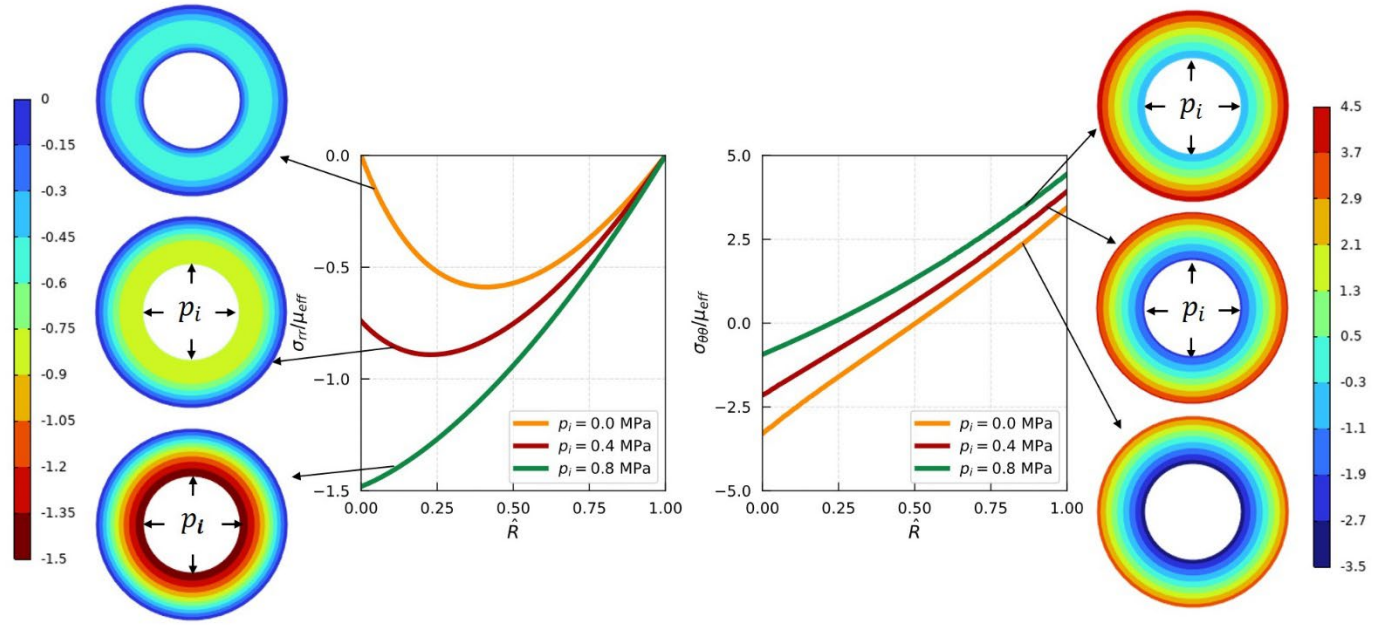


Figure 7. Effect of the internal pressure p_i on the stress response of LCE cylinders subjected to extension–torsion superimposed with inflation. The non-dimensional stress components σ_{rr}/μ_{eff} and $\sigma_{\theta\theta}/\mu_{eff}$ are plotted across the cylinder wall thickness for an inner-to-outer radius ratio of $R_i/R_o = 0.5$. The kinematic inputs are set as $\tau = 3\pi$ and $\lambda = 3.0$, respectively.

In addition to pure extension–torsion, inflation can be considered as part of a multiaxial loading scenario. Inflation is defined as the net pressure applied to the internal surface of the specimen, whether a cylinder or a balloon. In the present study, the outer surface is assumed to be traction-free, and the entire pressure load is applied to the inner surface, such that $\sigma_{rr}|_{r=r_i} = -p_i$.

Inflation has a negligible effect on the axial and shear stress components, $\sigma_{\theta z}$ and σ_{zz} , and primarily influences the radial and circumferential stresses, σ_{rr} and $\sigma_{\theta\theta}$. The stress components σ_{rr} and $\sigma_{\theta\theta}$ for LCE cylinders subjected to extension–torsion combined with inflation were evaluated for three different internal pressure levels. The corresponding results, together with two-dimensional schematics, are presented in Figure 7. The stress components obtained from the Anssari–Benam model are non-dimensionalized by the effective shear modulus, μ_{eff} , defined as $\mu_{eff} = 2 \left(\frac{\partial W}{\partial I_1} + \frac{\partial W}{\partial I_2} \right) \Big|_{I_1=3, I_2=3}$. Using the material parameters listed in Table 1, μ_{eff} is evaluated as 0.054 MPa.

As observed, σ_{rr} exhibits pronounced variation across the wall thickness, with the extremum shifting toward the inner radius as p_i increases. However, these variations decrease through the thickness and converge to zero at the outer traction-free surface of the cylinder ($R = R_o$). Furthermore, the magnitude of σ_{rr} increases with increasing internal pressure. This increase is more significant near the inner radius, while the differences between the curves diminish toward the outer surface.

One of the key parameters in soft materials with industrial applications is the magnitude of the forces and torques generated in response to environmental variables. For instance, in artificial muscles made from soft materials, the generated forces define the operating range by determining the muscle’s capability to push, lift, or hold objects when actuated. In this research, these generated forces and torques, referred to as resultant torque T and resultant force f , are evaluated using the following relations:

$$f = 2\pi \int_{r_i}^{r_o} \sigma_{zz} r dr, \text{ and } T = 2\pi \int_{r_i}^{r_o} \sigma_{\theta z} r^2 dr. \quad (19)$$

In Eq. (19), the stress components are integrated over the operational surface—the through-thickness cross-section of the cylinder—to yield the resultant force and torque. The variables r_i and r_o denote the current inner and outer radii of the cylinder, respectively.

Another important performance metric for soft-material devices is the blocking force, f_{block} , defined as the maximum force generated by the actuator under zero-displacement conditions. The blocking force quantifies the maximum load-bearing capacity of the actuator, with larger values indicating improved resistance to external loading. The axial blocking force of an LCE cylinder under torsional loading is defined as follows:

$$f_{block} = 2\pi \int_{r_i}^{r_o} \sigma_{zz} r dr \text{ at } \lambda = 1. \quad (20)$$

Equation (14) evaluates the blocking force by integrating the axial stress component over the operational surface of the cylinders when the axial displacement is constrained to zero ($\lambda = 1$).

Figure 8 showcases the magnitudes of the forces and resultant torques of the LCE cylinders as a function of kinematic and geometric parameters. The magnitude of the blocking forces (see Figure 8a) exhibits a monotonic, nonlinear increase with increasing τ for all geometric ratios, indicating increased axial stiffness of the cylinder as torsion progresses. Wall thickness dependency is observed in the magnitudes of the blocking forces: thicker cylinders (smaller R_i/R_o) generate significantly larger magnitudes, while thinner cylinders (larger R_i/R_o) exhibit reduced absolute forces. Negative values for the blocking force response under pure torsion demonstrate a reverse coupling between torsion and axial motion in the cylinders, causing them to shorten along their axial length when twisting. In the results, as τ increases, the divergence between the curves becomes more pronounced, indicating a nonlinear increase in the absolute blocking force values for all geometric ratios.

The resultant torques, shown in Figure 8b, increase nonlinearly with increasing τ for all geometric ratios. Similar to the blocking force trends, a reverse relationship between the geometric ratios and the magnitudes of the generated torques is observed. As the twist angle τ increases, the material exhibits softening and stiffening effects, respectively. This complex, non-monotonic behavior corresponds to the semi-softness—the stress-plateau region—and the limit of chain extensibility at moderate and high stretch levels, respectively. Such trends can also be observed in the LCEs' response shown in Figures 2 and 5. The trends seen for the resultant torques of LCEs subjected to extension-torsion are also repeated for the resultant forces, which are not reported here.

In Figure 8, the absolute values of the results demonstrate an exponential decrease as the ratio R_i/R_o increases for both resultant torques and blocking forces. This leads to a remarkable difference between the results for the cases with $R_i/R_o = 0.9$ and the rest of the cases, implying the significant role of the geometric ratio on the practical parameters of LCE cylinders used as soft actuators. It also reveals that there could be an optimum geometric ratio for soft materials with nonlinear responses in terms of their generated forces and torques.

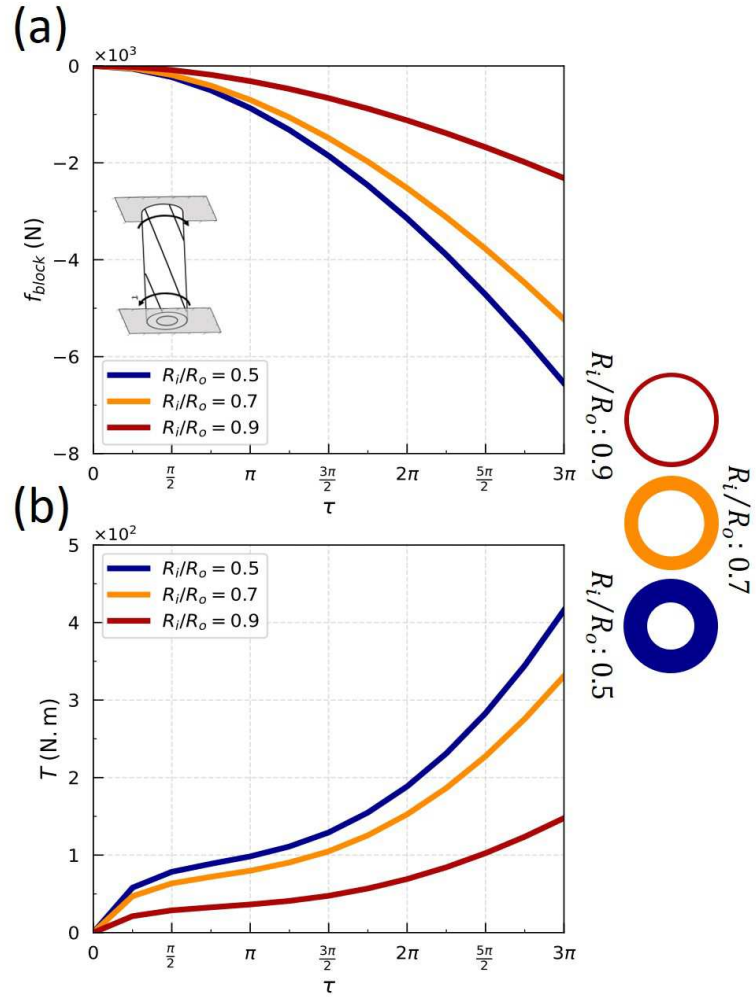


Figure 8. The influence of the geometric ratio R_i/R_o on (a) the blocking force f and (b) the resultant torque T of polydomain LCE cylinders under extension-torsion loading. The corresponding cross-sectional geometries are shown on the right for comparison.

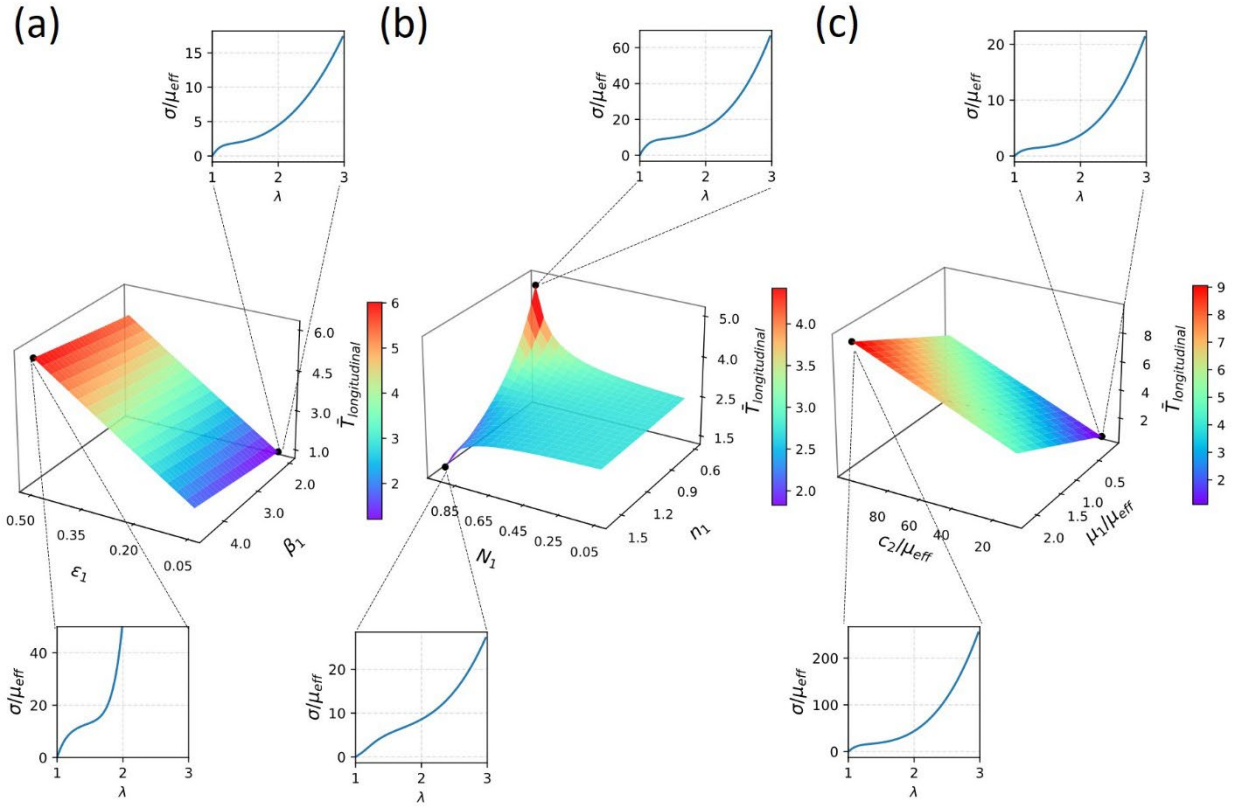


Figure 9. Influence of the Anssari–Benam model material parameters, namely (a) β_1 , ϵ_1 , (b) n_1 , N_1 , and (c) C_2 , μ_1 , on the resultant torque of a polydomain LCE cylinder subjected to combined extension–torsion loading. The stretch ratio λ and twist angle τ are fixed at 1.5 and $\pi/2$, respectively. The corresponding stress–stretch responses under uniaxial tension, associated with the maximum and minimum values considered in each parametric surface, are also presented.

Material parameters not varied in each panel are held constant at the reference values listed in Table 1.

We have previously introduced the Anssari–Benam model as a binomial constitutive framework capable of accurately capturing the behavior of LCEs under combined loading. We now examine its constituents in greater detail to clarify why the model performs effectively and how its material parameters influence the mechanical response. A parametric analysis of the non-dimensional Anssari–Benam model parameters and their effect on the non-dimensional resultant torque of a cylindrical specimen subjected to combined extension–torsion loading was conducted. The results are presented in Figure 10. The stretch ratio and twist angle were fixed at $\lambda = 1.5$ and $\tau = \pi/2$, respectively. For each parametric study, the remaining parameters—except for the one under investigation—were held constant at the reference values listed in Table 1.

The stress–stretch response of the Anssari–Benam model under uniaxial tension is also shown in Figure 10, corresponding to the maximum and minimum values considered in each parametric study, in order to illustrate the influence of material parameters on the mechanical behavior. As shown in the right panel of Figure 10, the material parameters C_2 and μ_1 exhibit a direct correlation with the resultant torque. Increasing either parameter leads to an approximately linear increase in mechanical stiffness. The parameters n_1 and N_1 display inverse and direct relationships with the output torque, respectively. Near $N_1 = 1$, sharp variations occur, as this value corresponds to a singularity of the model. In particular, N_1 governs the limiting extensibility of the polymer chains, as illustrated in the middle panel. Larger values of n_1 suppress the semi-soft elastic response, eliminating the characteristic plateau region at moderate stretch levels. Conversely, smaller values of n_1 slightly increase the material stiffness. Furthermore, the resultant torque is highly sensitive to variations in N_1 within a relatively

narrow parameter range. As shown in the left panel, the parameters β_1 and ε_1 both have a direct effect on the resultant torque, with larger values producing higher torque. The corresponding First Piola–Kirchhoff stress responses under uniaxial tension indicate that increasing β_1 and ε_1 accelerates the approach to the chain extensibility limit. In contrast, smaller values reduce the steepness associated with this limit and increase the extent of the plateau region at moderate stretches in the stress–stretch response. It should be noted that Figure 10 presents the resultant torque for LCE specimens. The same qualitative trends were observed for the resultant force, although the quantitative values differ.

3.2. Balloon response

Large deformation in LCE balloons is a particularly attractive phenomenon with potential applications in peristaltic pumping. An LCE balloon’s volume can increase up to 20 times its initial volume after applying a specific stimulus. To achieve a clear understanding of this phenomenon, we elaborate on the stress response of polydomain, thick-walled LCE balloons when subjected to inflation. Furthermore, the inflation-induced instability that occurs in thick-walled LCE balloons is discussed in detail. Before presenting the results, we verified the developed theoretical framework for LCE balloons (see Section 2.2) by comparing the stress response obtained from the analytical solution and direct Finite Element Method (FEM) analysis. For the FEM analysis, the balloon’s geometry was modeled as one octant (1/8) of a sphere, with symmetry constraints applied to the side surfaces to prevent numerical divergence when solving with the Newton-Raphson method. The verification results for the analytical and direct FEM analyses are shown in Figure 10 for two inner pressure levels. Both the auxiliary pressure (p) and the current radial coordinate across the wall thickness showed excellent agreement during verification. This agreement is maintained for both the FEM analysis and the analytical framework results before reaching high inner pressure thresholds. For $\hat{p}_i > 3$ (where $\hat{p}_i = p_i/\mu_{eff}$ serves as the non-dimensional inner pressure), the stress responses gradually diverge due to the occurrence of through-thickness instability.

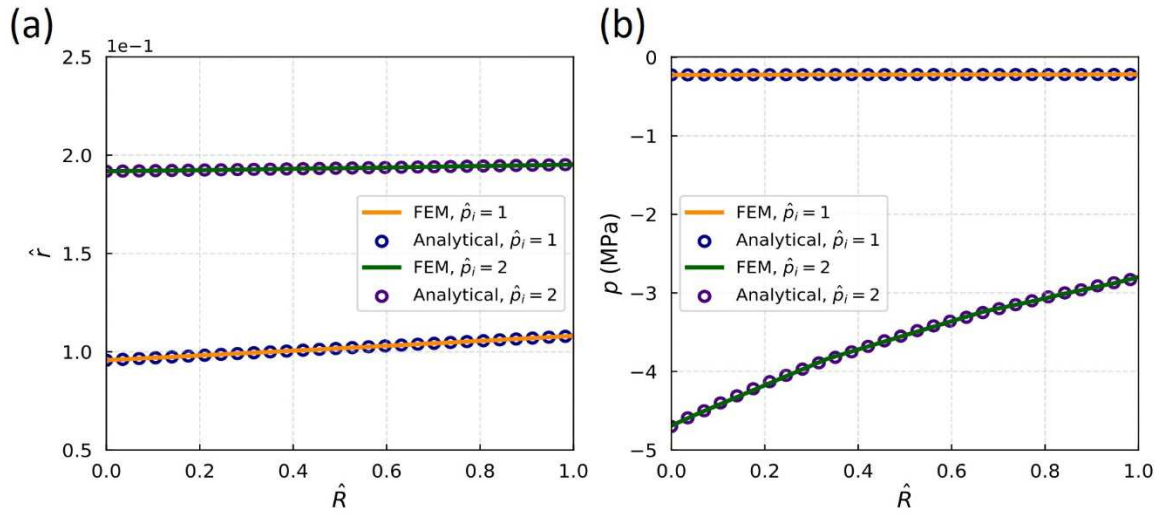


Figure 10. Analytical framework verification against FEM simulations for LCE balloons subjected to two inflation levels: (a) current radial coordinate across the balloon wall thickness plotted as a function of the non-dimensional arc length (initial radial coordinate); (b) auxiliary pressure p along the wall thickness (arc length). Solid lines represent the FEM results, while circles denote the analytical predictions.

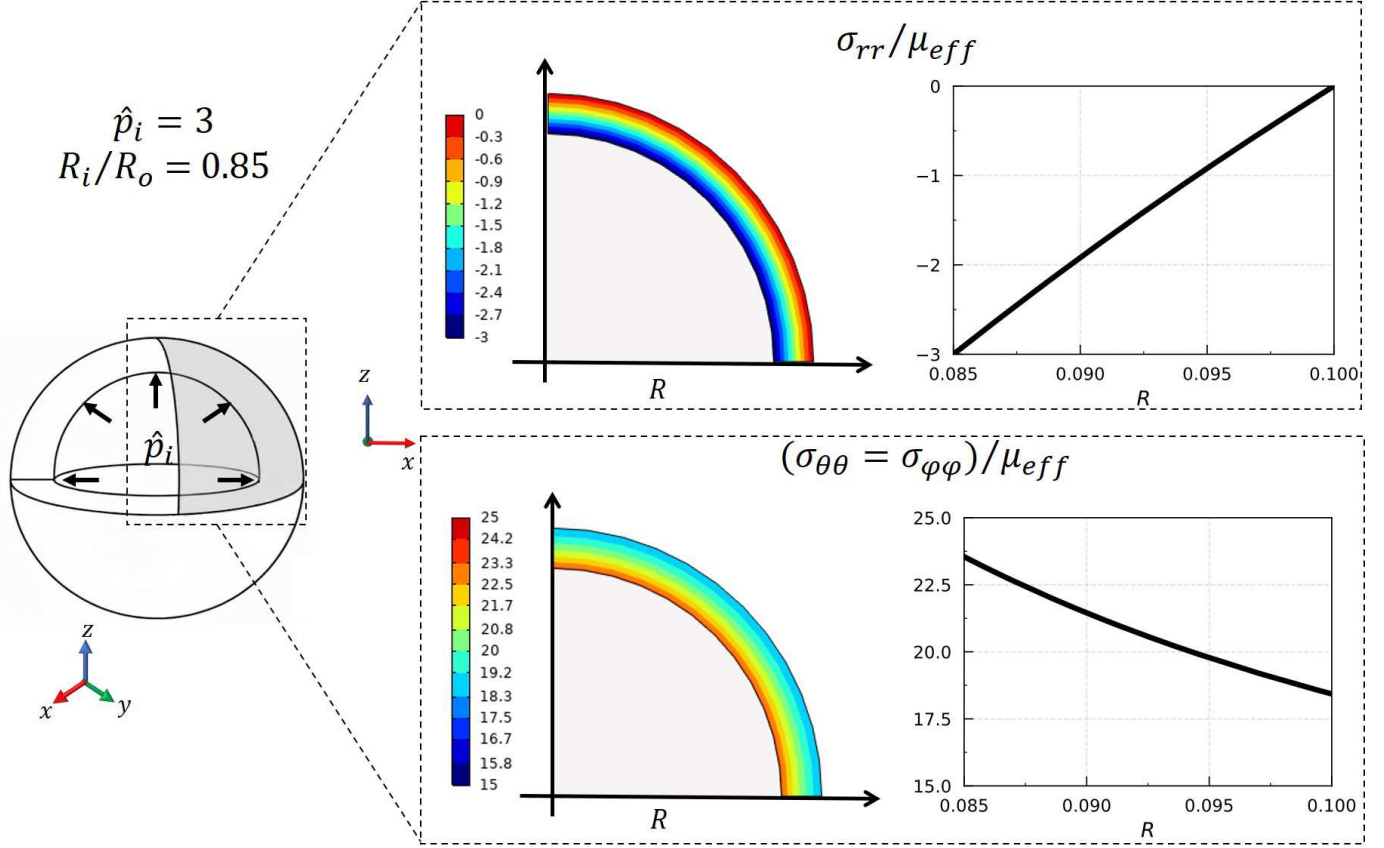


Figure 11. The distribution of normalized stresses along the wall thickness of the octant (1/8) of the balloon. Non-dimensional internal pressure and inner-to-outer radius ratio are fixed at $\hat{p}_i = 3$ and $R_i/R_o = 0.85$, respectively. The R -axis represents the initial radial coordinates.

The stress response of the LCE balloons under inflation indicates a dependency on the radial coordinate along the thickness. Figure 11 shows the distribution of non-dimensional stress components along the wall thickness axis, captured at sufficiently high-pressure values ($\hat{p}_i = 3$). Note that this figure was obtained by extending the contours of the 1D analytical solution onto a 2D surface. The radial stress component (σ_{rr}) shows a linear increase in value (a decrease in magnitude), approaching zero at the outer surface. The remaining circumferential stress components ($\sigma_{\theta\theta}$ and $\sigma_{\varphi\varphi}$) are equal in value because the inner pressure is applied homogeneously throughout the inner surface of the balloon, and the material (polydomain LCE) is also assumed to be isotropic. The distribution of $\sigma_{\theta\theta}$ and $\sigma_{\varphi\varphi}$ shows a gradual decrease along the thickness, with approximately a 20% relative change at the outer surface with respect to the inner one. The ratio of this variation also decreases along the thickness, demonstrating a nonlinear distribution for circumferential stresses.

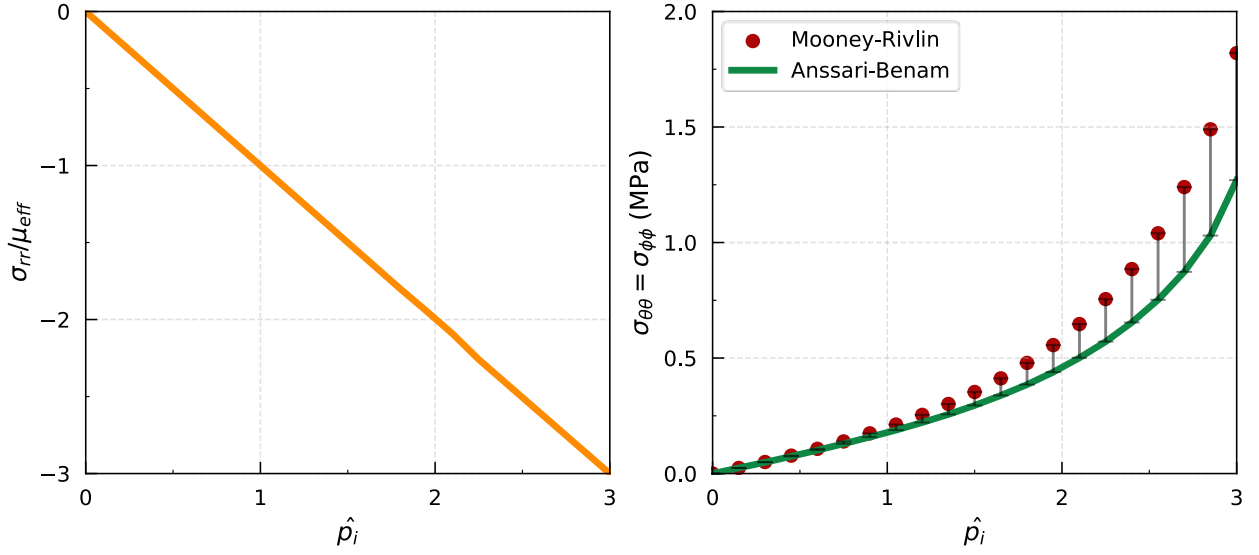


Figure 12. The analytical predictions of different constitutive models for $\sigma_{\theta\theta} = \sigma_{\phi\phi}$ (right panel) in an LCE balloon subjected to a linearly decreasing internal pressure (left panel). The inner-to-outer radius ratio is fixed at $R_i/R_o = 0.85$.

Similar to the LCE cylinders, we also assessed the stress response predictions of different constitutive models—specifically the Mooney-Rivlin and Anssari-Benam models—for LCE balloons under inflation. The predictions are presented in Figure 12, based on the material parameters listed in Tables 1 and 2. Results for the Yeoh strain energy function are omitted because it suffers from numerical divergence during the solution process due to the severe thickness variations. The non-zero stress components at the inner surface of the thick-walled LCE balloon ($R = R_i$) were evaluated assuming a linear increase in internal pressure (p_i). The magnitude of the radial stress component, σ_{rr} , is predicted to be equal to the magnitude of the applied inner pressure at each increment. Both strain energy functions predict identical values for σ_{rr} , as it serves as a boundary condition input for the problem. Note that σ_{rr} approaches zero as we move through the thickness toward the traction-free outer surface. As a result of the linearly increasing inner pressure (p_i), the hoop stress ($\sigma_{\theta\theta}$) and meridional stress ($\sigma_{\phi\phi}$) exhibit exponential increases, as demonstrated in the right panel of Figure 12. The predictions for $\sigma_{\theta\theta}$ and $\sigma_{\phi\phi}$ differ between the two models: the Mooney-Rivlin model predicts larger values than the Anssari-Benam model. This discrepancy increases with internal pressure, resulting in a divergence of approximately 44% in the Mooney-Rivlin prediction relative to the Anssari-Benam model at $\hat{p}_i = 3$.

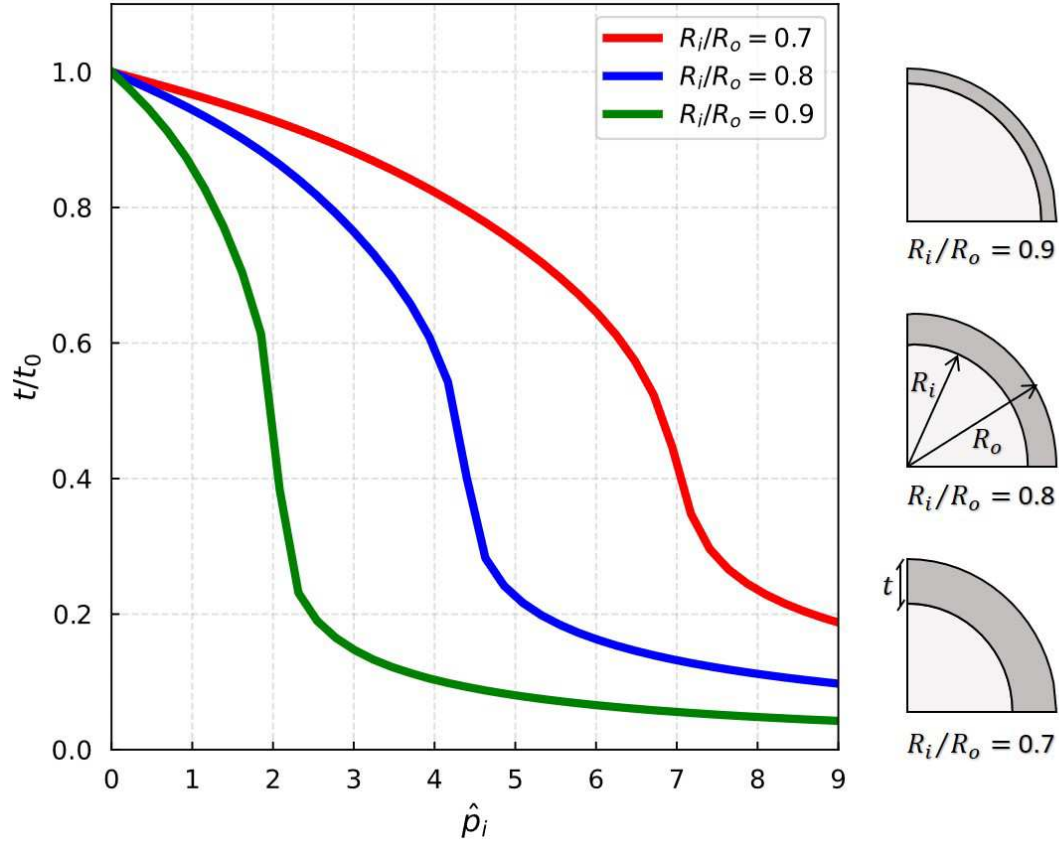


Figure 13. Through-thickness instability in LCE balloons under inflation, plotted as a function of the non-dimensional internal pressure for different inner-to-outer radius ratios R_i/R_o .

As previously mentioned, at sufficiently high internal pressures, through-thickness instability occurs in LCE balloons. Figure 13 illustrates the dimensionless variation of the LCE balloon wall thickness as a function of the applied internal pressure. For this analysis, the internal pressure was increased up to $\hat{p}_i = 9$ for balloons with three different initial thickness ratios ($R_i/R_o = 0.5, 0.7, 0.9$). The thicknesses shown on the y-axis are non-dimensionalized with respect to their initial values. In all plots, a pronounced and abrupt reduction in the balloon thickness is observed. This sudden thickness drop occurs at different pressure levels depending on the ratio R_i/R_o . As the initial balloon thickness decreases, the onset of this abrupt thinning shifts to lower internal pressures, and the magnitude of the thickness reduction becomes more severe. For very thin balloons, this sharp decrease in thickness may trigger geometric instability and can even lead to snap-through behavior. One possible explanation for this phenomenon lies in the semi-soft elastic response, an intrinsic characteristic of LCEs. Semi-softness refers to the enhanced compliance exhibited by LCEs at moderate stretch levels, arising from the rotation of a large fraction of mesogens toward the director, thereby reducing the energetic cost of deformation.

4. Conclusion

As a type of smart polymer, LCEs exhibit large reversible actuations that can be utilized in soft robotics, biomedical devices, and shape-morphing structures. This study presents a general theoretical framework for investigating polydomain LCEs under multiaxial loading. Utilizing isotropic hyperelasticity, this study

investigated the stress response of polydomain LCEs in thick-walled cylindrical and balloon-like structures. The cylinders were subjected to extension–torsion and extension–torsion–inflation, while the balloons were subjected to inflation. Three different forms of the strain energy function were adopted, and their respective aspects were discussed in detail to provide a comprehensive understanding of the characteristics of different constitutive models and to emphasize the critical role of model selection in such problems. This study suggests choosing strain energy functions of binomial form, which are capable of capturing material response up to high stretch values during calibration, to achieve the most robust predictions. For instance, up to a 39.2% difference was reported in the radial stress component predictions between binomial and non-binomial models. The binomial strain energy function adopted in this study, termed the Anssari-Benam model, was also discussed in detail, and the role of its material parameters was highlighted. This study also evaluated blocking force and longitudinal torque in relation to geometric parameters for LCE cylinders, considering them as critical performance metrics for soft actuators. Moreover, the role of wall thickness was assessed regarding the occurrence of through-thickness instability in LCE balloons. The theoretical framework developed, along with its findings, provides a clear and practical approach for analyzing and designing polydomain LCE systems, particularly for applications in soft robots and shape-adaptive structures. Future studies could incorporate temperature-dependency and anisotropy into the model to make it better suited for nematic LCEs.

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Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

A.1 Cauchy stress components for the cylinder:

$$\begin{aligned}\sigma_{rr} &= -\frac{2 C_\varepsilon \varepsilon_k \left(\frac{\bar{I}_2}{3}\right)^{\varepsilon_k}}{\bar{I}_2 \left(\frac{dr(R)}{dR}\right)^2} + \frac{3N\mu(n-1) \left(-\frac{\beta}{\bar{I}_1 - 3N} (\bar{I}_1 - 3N) + \frac{\beta(\bar{I}_1 - 3)^\beta}{3N(n-1)(\bar{I}_1 - 3)} \right) \left(\frac{dr(R)}{dR} \right)^2}{n} - p \\ \sigma_{\theta\theta} &= -\frac{2 C_\varepsilon R^2 \varepsilon_k \left(\frac{\bar{I}_2}{3}\right)^{\varepsilon_k}}{\bar{I}_2 r^2(R)} + \frac{3N\mu(n-1) \left(-\frac{\beta}{\bar{I}_1 - 3N} + \frac{\beta(\bar{I}_1 - 3)^\beta}{3N(n-1)(\bar{I}_1 - 3)} \right) \left(\tau^2 \gamma^2 r^2(R) + \frac{r^2(R)}{R^2} \right)}{n} - p \\ \sigma_{\theta z} &= \frac{2 C_\varepsilon R^2 \varepsilon_k \tau \left(\frac{\bar{I}_2}{3}\right)^{\varepsilon_k}}{\bar{I}_2 r(R)} + \frac{3N\gamma^2 \mu \tau (n-1) \left(-\frac{\beta}{\bar{I}_1 - 3N} + \frac{\beta(\bar{I}_1 - 3)^\beta}{3N(n-1)(\bar{I}_1 - 3)} \right) r(R)}{n} \\ \sigma_{zz} &= -\frac{2 C_\varepsilon \varepsilon_k \left(\frac{\bar{I}_2}{3}\right)^{\varepsilon_k} \left(\tau^2 R^2 + \frac{1}{\gamma^2} \right)}{\bar{I}_2} + \frac{3N\gamma^2 \mu (n-1) \left(-\frac{\beta}{\bar{I}_1 - 3N} + \frac{\beta(\bar{I}_1 - 3)^\beta}{3N(n-1)(\bar{I}_1 - 3)} \right)}{n} - p\end{aligned}$$

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